Final Project Report

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BTMA 431: Gathering, Wrangling and Analyzing Data in R

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1.Introduction: Air quality is an important environmental factor that directly affects daily life, health, and outdoor activities. During my time in Canada, I realized that air-quality information is monitored much more frequently than what I was used to in Vietnam. As an international student coming from a tropical and relatively stable climate, I immediately noticed how quickly weather conditions shift in Canadian cities and how often the air quality is updated on public screens, transit systems, and weather platforms. Vietnam does not use the Air Quality Health Index system, so my initial exposure to AQHI came only after arriving in Canada. This raised my curiosity about how it works and why it varies so much across different regions.

The Air Quality Health Index is a Canadian measurement that communicates the health risks associated with local air pollution. Instead of describing pollution in technical or chemical terms, AQHI simplifies the concept into a numeric scale that reflects how air quality may influence human health. The index is calculated using concentrations of three major pollutants including ground-level ozone, particulate matter, and nitrogen dioxide. It converts these pollutants into a single number ranging from low to very high risk. This system is widely used in Canadian cities because of the changing climate, seasonal wildfire activity, traffic emissions, and geographic effects that can influence pollution levels. Learning about AQHI helped me understand why air quality behaves differently here compared to Vietnam and why it is important to monitor these changes in everyday decision-making.

Recognizing these differences motivated me to explore what actually drives the fluctuations in AQHI. I wanted to understand the broader picture: whether changes come mainly from weather conditions, from the characteristics of a city's location, or from the timing of environmental patterns throughout the day. Rather than viewing AQHI as a single output, I became interested in the underlying factors that shape it and how they interact across different regions of Canada. This project is therefore built around the idea that several environmental and temporal components contribute to air-quality patterns, and analyzing them together can provide a more complete understanding of how AQHI behaves.

To structure the study, I focused on three primary research questions closely connected to these real-world influences. The first examines how key weather variables such as temperature, humidity, wind speed, and precipitation relate to AQHI, and which of these variables shows the strongest relationship. The second compares inland and coastal cities to determine whether geographic location plays a measurable role in air-quality differences, since coastal regions tend to have stronger air circulation while inland regions may trap pollutants more easily. The third investigates whether AQHI levels vary between weekdays and weekends and how these differences change across different parts of the day, including morning, afternoon, evening, and night, in order to understand the time-based patterns that influence air quality. Using weather-condition data from the Open-Meteo API and AQHI records from Government of Canada monitoring stations, this project combines five cities representing different regions: Calgary, Vancouver, Victoria, Toronto, and Ottawa. The analysis includes data wrangling, visualization, confidence-interval estimation, hypothesis testing, and multiple regression models. By examining the results across the three research areas, the project aims to provide a clearer understanding of how weather, location, and daily behavior influence AQHI across Canadian cities.

Overall, this report reflects both an academic inquiry and a personal learning experience. As someone adapting to a new climate and environmental system, I wanted to understand not only how AQHI works but also what patterns lie beneath it. The findings from this analysis help illustrate the dynamics of air quality in Canada and offer meaningful insights into the environmental factors that shape the index that Canadians rely on every day.

2. Methodology

2.1 Data Sources: This project relies on two main sources of information: hourly Air Quality Health Index (AQHI) measurements and hourly weather data. Both types of data were collected for the same five Canadian cities, namely Calgary, Vancouver, Toronto, Victoria, and Ottawa. To make the analysis meaningful, all information was gathered for a consistent period, specifically from October 20, 2025 at 01:00 UTC until October 27, 2025 at 00:00 UTC, ensuring that every city shared the exact same timeline.

The AQHI information came from the Government of Canada's Open Data Portal, which publishes hourly air-quality observations recorded at official monitoring stations. Each city has several stations, and the raw CSV files list these stations in separate columns using their unique codes. Because the objective of this project is to examine air quality at the city level rather than at the station level, I selected one representative monitoring station for each city. The stations used in the analysis were IAKID for Calgary, JAZBU for Vancouver, FDQBU for Toronto, FEVNT for Ottawa, and JCLMX for Victoria. These codes are embedded directly in the raw files, so the data were reshaped into a long format that linked each timestamp to the AQHI reading from the chosen station. The date and hour fields were combined into a single datetime variable, converted into UTC, and prepared for merging with weather observations.

Weather data were collected through the Open-Meteo API, a free service that provides historical weather records at an hourly resolution. The API retrieves data based on geographic coordinates, so each city was defined by its latitude and longitude. This allowed the API to return localized conditions for each location. Four variables were included in the dataset because of their known relevance to air-quality behavior: temperature at two meters, relative humidity at two meters, precipitation, and wind speed at ten meters. All values were downloaded with timestamps already aligned to UTC, which made it straightforward to match them with the AQHI records.

Once both datasets were collected, the weather data and the AQHI readings were matched using the city name and the exact UTC timestamp. For the analysis, I used only the inner join version of the merged dataset. This ensured that every observation included complete information for all variables such as AQHI, temperature, humidity, wind speed, and precipitation at the same point in time. Working with a fully matched dataset was important because it allowed the statistical tests and regression models to be based on reliable and consistent data. The AQHI values came from the chosen representative monitoring station for each city, which helped maintain consistency while avoiding unnecessary loss of data from stations that had missing readings.

Together, these steps produced a clean, synchronized dataset that forms the basis for every visualization, statistical test, and regression model used throughout the study.

2.2 Data Cleaning and Processing : The cleaning process began with preparing the weather dataset. I first created a table of the five selected Canadian cities and their corresponding latitude and longitude coordinates. These coordinates were used to request hourly weather information from the Open-Meteo API. For each city, I sent the same query covering the UTC time range from October 20 to October 27, 2025, and retrieved four key variables: temperature, relative humidity, precipitation, and wind speed. The data returned in JSON format were converted into a tibble with fromJSON, and I constructed a unified datetime field while keeping only the variables necessary for analysis. The resulting weather dataset was already clean, consistently structured, and ready to merge.

The AQHI data required more substantial preprocessing. The raw government files were in a wide format, with one date column, one hour column, and multiple AQHI columns representing different monitoring stations. To handle this structure, I wrote a function that automatically identifies the date and hour fields, reshapes the data into long format, and creates a proper UTC datetime stamp for each row. During this step, I converted AQHI values to numeric form because the raw files sometimes contain non-numeric entries. I applied suppressWarnings during conversion to avoid interruptions from minor formatting inconsistencies.

After both datasets were cleaned independently, the final step was to merge them using the city name and the exact UTC timestamp. I used an inner join because it keeps only the rows where weather readings and AQHI values are recorded at the same moment in time. This ensured that every observation in the merged dataset contained complete information for all variables used in the study.

Only after the merge was completed did I remove missing values. This step was important because AQHI files sometimes contain gaps for certain hours, and the weather API occasionally reports missing precipitation or wind values. Dropping NA values after merging allowed me to keep as much valid data as possible during the joining process, and then remove only the rows that were truly incomplete. By doing this, the final dataset became fully synchronized and suitable for regression modeling, statistical tests, and visualization without the risk of partial observations affecting the results.

2.3 Statistical Methods: The analysis relied on a combination of statistical techniques designed to uncover relationships between AQHI and several environmental and categorical factors. The first set of methods focused on understanding how AQHI responds to changes in weather conditions. To evaluate these relationships, I used correlation analysis to measure the strength and direction of linear associations between AQHI and variables such as temperature, humidity, wind speed, and precipitation. This provided an initial sense of which weather factors tended to move alongside AQHI before building more formal models.

To quantify these effects more precisely, multiple linear regression was used as the primary modeling approach for Question 1. The baseline model predicted AQHI using all four weather variables. A second version of the model included city fixed effects to account for regional differences. I also estimated a standardized regression model, where all variables were transformed into z-scores, allowing the effect sizes of different weather factors to be directly compared on a single scale. This helped determine which weather variable had the strongest influence on AQHI.

Regression analysis was also applied to the categorical comparisons in Question 2 and Question 3. For Question 2, a simple linear regression model was used to examine whether inland or coastal cities tend to have higher AQHI values. This was done by treating region type as a categorical predictor and estimating how much AQHI changes when moving from a coastal to an inland location, while keeping other factors constant. In the follow-up portion of Question 2, I extended this idea by fitting a model that used the city itself as the predictor. This allowed the analysis to quantify how AQHI differs across specific cities after accounting for city-level patterns.

A similar regression approach was used for Question 3. To test whether AQHI differs between weekdays and weekends, I fit a model with day type as the predictor. This estimated the average change in AQHI when the day shifts from a weekday to a weekend. Because the analysis later explored time-of-day effects as well, the regression served as a clean baseline measure of the overall weekday–weekend difference before breaking the patterns down by specific time periods.

Beyond regression, the study also incorporated several inferential and descriptive techniques. Two-sample t tests were used to determine whether differences between groups—such as inland versus coastal cities, or weekdays versus weekends—were statistically significant and not due to random variation. Summary statistics such as means, standard deviations, and 95 percent confidence intervals supported these comparisons by showing how AQHI levels varied within each group.

Visualization played a central role throughout the analysis. Scatterplots with regression lines were used to illustrate weather–AQHI relationships from Question 1. Heatmaps visualized correlation patterns, while bar charts displayed standardized effect sizes. Density plots, histograms, ridge plots, and confidence-interval bar charts helped compare AQHI distributions across cities, regions, and day types. For Question 3 follow-up, line charts and difference charts illustrated how AQHI changes across morning, afternoon, evening, and night between weekdays and weekends.

Altogether, these statistical methods allowed the project to examine AQHI from multiple angles, moving from simple comparisons to more structured regression modeling, and finally to visual patterns that helped interpret how AQHI behaves across weather conditions, geographic locations, and daily time cycles.

3. Result

3.1 Weather Condition: To address the first research question, I first estimated a multiple linear regression model to see how hourly AQHI responds to changes in temperature, humidity, wind speed, and precipitation. The model is statistically strong overall, with an F-statistic of 187.2 and a p-value below 0.001. The R-squared of 0.48 indicates that these weather factors jointly explain almost half of the variation in AQHI. (see Appendix A.1)

Looking at individual coefficients, temperature shows a small but significant positive effect, meaning AQHI tends to increase slightly as temperature rises. Relative humidity has a large and highly significant negative coefficient, making it the strongest predictor in the model; higher humidity is consistently associated with lower AQHI. Wind speed does not appear to play a meaningful role, as its p-value is

well above 0.05. Precipitation has a modest positive effect and reaches significance, suggesting AQHI increases slightly during hours with rainfall.
After preparing the merged dataset and ensuring all variables were complete, the analysis proceeded using standardized weather features and linear regression to quantify the relationship between AQHI and temperature, humidity, wind speed, and precipitation.

To understand how these variables relate to each other, I calculated a correlation matrix using the numeric weather features. This provided an initial overview of whether temperature, humidity, wind speed, and precipitation tend to move together or in opposite directions, helping to contextualize later findings.
Next, I estimated a series of regression models. The first model used only the weather variables to predict AQHI, while the second model added city fixed effects to control for systematic differences across locations. To compare the relative importance of each weather factor, I standardized all variables and re-ran the model on z-scores. This allowed the coefficients to be interpreted on the same scale, making it easier to see which predictors had the strongest influence on AQHI.
The resulting standardized estimates were stored in a summary table (imp), which was then used to support both the text interpretation and the visualizations presented in the next section.

To explore the relationship between AQHI and each weather variable, I created a set of scatterplot–regression charts. The plotting code first maps the weather variable to the x-axis, AQHI to the y-axis, and uses city as the color grouping. Each chart overlays a linear smoothing line to highlight the overall trend across observations.

The points are shown with reduced opacity so overlapping values remain visible, and the minimal theme keeps the visual focus on the patterns rather than styling elements. The same structure is repeated for temperature, humidity, wind speed, and precipitation, with only the x-variable changed in each plot.

The four scatterplots illustrate how AQHI varies with different weather conditions across cities. Temperature shows a consistent positive relationship with AQHI, with Calgary and Ottawa exhibiting the strongest upward trends, while Vancouver, Victoria, and Toronto show smaller but still positive slopes (see Appendix A.2). Relative humidity displays the opposite pattern: AQHI decreases as humidity rises, and this negative association is especially pronounced in Calgary and Ottawa. These patterns align with the regression model, where humidity had the largest negative and statistically significant coefficient (see Appendix A.4). Wind speed shows weak and inconsistent effects across cities—some lines slope slightly upward, others remain nearly flat—reflecting the regression result that wind speed was not a significant predictor (see Appendix A.3). Precipitation also exhibits minimal influence on AQHI; most data points occur on days without rainfall, and the fitted lines have very small slopes, matching the model's finding of a small but statistically detectable effect (see Appendix A.5). Overall, the visualizations support the regression estimates, confirming that temperature and humidity are the primary weather variables associated with changes in AQHI, whereas wind speed and precipitation play a comparatively minor role.

The correlation heatmap shows that temperature has the strongest positive correlation with AQHI ($r \approx 0.25$) among all weather variables, while relative humidity is moderately negative ($r \approx -0.68$) and wind speed/precipitation remain weak (see

Appendix A.6). These patterns are confirmed in the standardized regression results, where temperature has by far the largest standardized beta, indicating it is the dominant predictor of AQHI after accounting for all variables simultaneously. The per-city temperature plots further support this pattern: every city shows a clear upward trend where AQHI increases as temperature rises, though the slope varies (Ottawa and Victoria show especially strong effects, while Calgary has a milder slope) (see Appendix A.7). Finally, the PowerBI TempBucket visualization reinforces the relationship from a categorical perspective: higher temperature buckets (20–30°C) consistently correspond to higher median AQHI, especially in Vancouver and Victoria (see Appendix A.8). Together, these visualizations provide consistent evidence that temperature is the weather factor most strongly associated with increased AQHI across all five cities, while humidity, wind, and precipitation have weaker and more mixed influences.

3.2 Location Factor: This code first creates a simple mapping that classifies each city as either Coastal (Vancouver, Victoria) or Inland (Calgary, Toronto, Ottawa). It then adds this new variable (region_type) into the dataset and groups the data to calculate basic statistics—mean, median, standard deviation, and sample size for AQHI in each region. Finally, it performs an independent-samples t-test to formally compare whether mean AQHI differs between Coastal and Inland regions, and returns a tidy summary of the test.

The two-sample Welch t-test compares the mean AQHI of Coastal cities (Vancouver, Victoria) and Inland cities (Calgary, Toronto, Ottawa). The results show a large and highly significant difference between the two groups. Inland cities have a higher mean AQHI (1.92) than coastal cities (1.60). The test statistic is $t = -8.97$ with df ≈

786, and the p-value < 2.2e-16, which is far below any conventional significance threshold. The 95% confidence interval for the difference in means (Coastal – Inland) ranges from –0.388 to –0.249, meaning the inland mean is consistently higher across all plausible values. (see Appendix B.1)

Hypothesis Test

H_0 (Null Hypothesis):

The mean AQHI is the same for Coastal and Inland regions ($\mu_{(Coastal)} = \mu_{(Inland)}$).

H_1 (Alternative Hypothesis):

The mean AQHI differs between the two regions ($\mu_{(Coastal)} \neq \mu_{(Inland)}$).

Because the p-value is extremely small (< 2.2e-16), we reject H_0.

There is strong statistical evidence that inland cities experience higher AQHI levels than coastal cities.

To complement the t-test, I also ran a simple linear regression with AQHI as the outcome and region type (Coastal vs. Inland) as the predictor. The intercept represents the estimated mean AQHI for Coastal cities, which is 1.60, matching the descriptive statistics. The coefficient for region_typeInland = 0.319 indicates that Inland cities have, on average, 0.32 higher AQHI than Coastal cities. This effect is highly statistically significant (t = 7.71, p = 3.84e-14), confirming that the regional difference is strong and unlikely due to chance (see Appendix B.2). The overall model is simple but reinforces the same conclusion as the t-test: Inland regions consistently exhibit higher AQHI levels compared to Coastal regions.

The histogram shows that coastal cities (Vancouver and Victoria) have AQHI values tightly concentrated around 1.5 – 2.0, while inland cities (Calgary, Ottawa, Toronto) span a much wider range and frequently reach 2.5 – 3.5, indicating poorer air quality (see Appendix B.3). This pattern becomes even clearer in the density plot, where the coastal curve forms a sharp peak at low AQHI values, whereas the inland curve shifts to the right and displays a long tail, reflecting more frequent moderate-to-high AQHI levels (see Appendix B.4). Together, the two visualizations illustrate the same conclusion from different angles: coastal regions consistently exhibit cleaner air, while inland regions experience higher and more variable AQHI levels.

To examine whether AQHI levels differ systematically across cities, I estimated a regression model using city as a categorical predictor. Calgary serves as the baseline city, so each coefficient represents how much lower (or higher) the average AQHI is in the other cities relative to Calgary (see Appendix B.5). The results show that all four cities have significantly lower AQHI than Calgary, with coefficients around –1.00 to –1.20 and extremely small p-values ($p < 2e\text{-}16$). This means that, holding everything else constant, Calgary consistently reports the highest AQHI, while Ottawa, Toronto, Vancouver, and Victoria all show substantially cleaner air. The high F-statistic and $R^2 \approx 0.59$ indicate that the city alone explains a large portion of the variation in AQHI, reinforcing that geographic location has a strong influence on air quality even beyond coastal and inland grouping.

The four visualizations collectively show how AQHI levels differ across cities and regions. The mean AQHI bar chart with 95% confidence intervals highlights that Calgary consistently reports the highest average AQHI, while Ottawa has the lowest, and the tight confidence intervals indicate these differences are statistically meaningful (see Appendix B.6). The city-level density plots reinforce this pattern by showing Calgary's distribution shifted farther to the right, meaning higher AQHI readings overall, whereas coastal cities like Vancouver and Victoria cluster more tightly around lower values (see Appendix B.7). The treemap of median AQHI by city then summarizes these patterns in a compact way: Calgary again stands out with the largest block and highest median, while Ottawa and Toronto occupy smaller blocks, reflecting cleaner inland air (see Appendix B.8). Finally, the time-series plot of median AQHI by region type shows how coastal and inland cities evolve over time, with inland areas generally exhibiting higher and more variable AQHI values, especially in earlier periods (see Appendix B.9). Together, these charts tell a consistent story: AQHI varies substantially by city, with inland locations—especially Calgary—showing worse air quality than coastal regions.

3.3 Time condition: To study whether air quality differs between weekdays and weekends, I first built a focused dataset df3 containing only datetime, city, and AQHI, and removed rows with missing AQHI values. I then created a new categorical variable day_type, which labels each observation as Weekday or Weekend based on the calendar day, and stored it as an ordered factor (Weekday first, Weekend second). Next, I computed descriptive statistics and 95% confidence intervals for AQHI within each city × day_type combination (sample size, mean, standard deviation, standard error, and CI bounds), giving a detailed table of how average AQHI changes by city and by day type. Finally, I ran two sets of t-tests: an overall test comparing weekday vs weekend AQHI across all cities (tt_overall), and separate t-tests within each city (tt_city) to see whether the weekday–weekend difference is statistically significant at the city level.

Two sets of t-tests were performed to examine whether AQHI differs between weekdays and weekends. The overall t-test shows no significant difference (p =

0.569), with mean AQHI changing only slightly from 1.78 on weekdays to 1.81 on weekends (difference ≈ 0.024) (see Appendix C.1). However, the city-level t-tests reveal strong differences in several cities. For example, Calgary's weekend AQHI is 0.24 higher than weekdays ($t = 4.41$, $p = 2.0e-5$), and Ottawa shows an even larger difference of 0.37 ($p = 1.0e-9$). Vancouver and Victoria show the opposite pattern, with lower weekend AQHI (e.g., Vancouver = –0.25, $p = 2.9e-12$) (see Appendix C.1). This indicates that while the national pattern appears flat, meaningful weekday–weekend effects emerge once results are broken down by individual cities.

The regression model testing whether AQHI differs between weekdays and weekends shows no meaningful effect. The weekend coefficient is –0.025, meaning weekend AQHI is only about 0.02 lower than weekday levels on average — essentially no change. The effect is statistically insignificant ($p = 0.604$), and the model explains almost no variation in AQHI ($R^2 \approx 0.0003$) (see Appendix C.2). This confirms that, at the national level, weekday–weekend differences are minimal, matching the results from the overall t-test.

To visualize these weekday–weekend differences more clearly, I examined two sets of charts that compare AQHI patterns across cities. The boxplots provide a fuller picture of the AQHI distribution by showing spread and outliers. Calgary again stands out with higher AQHI ranges during both weekdays and weekends, but the median only drops slightly on weekends. Ottawa and Toronto have low and tightly clustered AQHI values with minimal weekday–weekend variation. Vancouver and Victoria display slightly higher weekend medians, but the boxes overlap substantially, reinforcing that the difference is small (see Appendix C.3). Overall, the boxplots confirm that AQHI varies more by city than by day type, consistent with the t-test and regression results. The bar chart comparing mean AQHI between weekdays and weekends across cities shows that the differences are small and city-specific. Calgary has the highest overall AQHI, and its weekday value (≈2.7) is slightly higher than weekend (≈2.4). Ottawa and Toronto show very minor drops on weekends, while Vancouver and Victoria display the opposite pattern, with weekend AQHI slightly higher (see Appendix C.4). The error bars (95% CI) overlap heavily in every city, which visually aligns with the statistical results showing that weekday–weekend differences are not significant overall.

The individual-points plot shows all AQHI observations for weekdays and weekends. The spread of the dots is almost the same in both groups, with most values clustered around AQHI ≈ 1.5–2.5. There is no clear shift upward or downward for weekends, which visually matches the statistical results showing no significant overall difference (weekday mean ≈ 1.81 vs weekend mean ≈ 1.78) (see Appendix C.5). The violin plot gives a smoothed view of the distribution and includes a boxplot for each day type. Both violins have almost identical shapes, with similar medians (around 1.6–1.7) and similar upper and lower ranges (see Appendix C.6). The overlap is nearly complete, reinforcing that weekday and weekend AQHI distributions are essentially the same when looking across all cities combined. To further understand whether AQHI varies across different times of day, I examined several visualizations that compare weekday and weekend patterns across cities. The line chart shows how AQHI changes across the afternoon, evening, morning, and night for each city. Calgary, Ottawa, Toronto generally have higher weekday AQHI, while cities like Vancouver, Victoria show slightly higher weekend values at certain times, especially morning and night (see Appendix C.7). Overall, the patterns fluctuate but the gap between weekday and weekend is small. The difference chart directly compares the two day types. Calgary, Ottawa, Toronto have negative values, meaning AQHI is lower on weekends. In contrast, Vancouver and Victoria show positive differences in several time blocks, which indicates higher weekend AQHI in those periods (see Appendix C8). This chart clearly highlights that the weekday–weekend effect varies by region. The ridgeline density plot shows how AQHI distribution shifts across key time periods: morning commute, late night, and evening. The weekday and weekend curves largely overlap, suggesting only small differences in AQHI between weekdays and weekends. However, some periods, particularly in the morning, show slight variations in the distribution. (see Appendix C.9). The circular chart summarizes the mean AQHI across four daily segments. It shows that AQHI tends to be higher during weekdays in the morning and night, while weekends see some slight increases during the morning and evening periods. The clock layout effectively highlights these daily variations and complements the observations from the previous charts. (see Appendix C10). Overall, the results show that AQHI does change across different times of day, but the weekday–weekend gap is generally small and highly city dependent. Inland cities tend to have slightly higher AQHI on weekdays, especially in the afternoon and evening, while coastal cities often show higher weekend levels in the morning and night.

3.4 Overall Result: The analysis across weather conditions, location (coastal vs inland), and time of day provided valuable insights into the variation of AQHI. Weather conditions, particularly temperature and humidity, had the most significant impact on AQHI levels, with temperature showing the strongest relationship. Coastal cities generally had more stable air quality, while inland cities tended to experience higher AQHI levels, likely due to the accumulation of pollution. Time of day also influenced AQHI patterns, with weekdays exhibiting more variation in air quality compared to weekends, especially in inland cities. Overall, the findings emphasize the complex interplay between environmental conditions, geographic location, and daily activities, underscoring the need for a multi-faceted approach to understanding air quality trends across different regions.

4. Discussion: This project examined the impact of weather, location, and time of day on the Air Quality Health Index (AQHI) across five major Canadian cities: Calgary, Vancouver, Toronto, Victoria, and Ottawa. By addressing three main research questions, we found that weather conditions, particularly temperature, have the most significant impact on AQHI. Higher temperatures tend to correlate with poorer air quality, especially in inland cities. Coastal cities, with more consistent airflow and cooler temperatures, generally experience better air quality compared to inland cities, where pollution accumulates more easily.

When analyzing AQHI variations across weekdays and weekends, I found that the differences were minimal overall, with slight variations observed during the morning and night in coastal cities. These changes were likely due to higher human activity during weekends, particularly in urban areas. The time-of-day analysis highlighted how AQHI varies during specific periods, but overall, the findings showed that environmental and temporal factors have a measurable effect on air quality.

This research offers valuable insights for international students, immigrants, and tourists considering living in or visiting Canada. For international students, understanding AQHI variations across different cities can help in making more informed decisions about where to study, especially for those with respiratory issues. While AQHI is just one of many factors to consider, it provides a tangible metric for comparing cities based on air quality. Additionally, the findings can assist newcomers in choosing cities with healthier environments, and even tourists can use this data to plan visits with health considerations in mind.

Beyond personal decisions, this research has significant implications for urban planning and policy-making. City officials and urban developers can use these findings to better understand how environmental conditions affect AQHI, enabling them to develop strategies to improve air quality. This could include initiatives such as reducing emissions, increasing green spaces, and improving public transportation. Moreover, this study highlights the need for further research into the long-term effects of air quality on health and the potential benefits of targeted interventions to reduce pollution in high-risk areas.

While this study provides valuable insights into how weather, location, and time of day affect AQHI, future research could delve deeper into the influence of traffic patterns and human behavior on air quality. For instance, studying the impact of rush hour traffic, commuting patterns, and urban congestion could help identify specific periods when AQHI spikes, offering opportunities for targeted interventions. Additionally, exploring the effects of lifestyle factors, such as indoor activities, use of public transport, and outdoor recreational habits, could provide a more holistic understanding of how individuals’ actions contribute to air quality. Further research into these areas could enable more precise predictions of AQHI fluctuations, which would support more effective urban planning and environmental policies.

While the analysis provides meaningful insights into AQHI patterns, several limitations should be acknowledged. The dataset covers only a single week, which may not fully represent seasonal variation or extreme events such as wildfires or winter inversions. AQHI values were taken from one monitoring station per city, meaning within-city air quality differences were not captured. Additionally, human factors such as traffic volume, industrial schedules, or special events were not included in the models. These constraints suggest that the findings should be interpreted as short-term patterns rather than long-term generalizations.

In conclusion, while AQHI is just one factor in the decision-making process, it offers valuable insights into the environmental conditions of Canadian cities. By understanding how AQHI varies by weather, location, and time of day, individuals, policymakers, and planners can make more informed choices that contribute to improving urban living environments and public health.

References

Government of Canada. (2025). Air Quality Health Index observations (AQHI): https://open.canada.ca/data/en/dataset/28936e1b-681f-4c73-b04a-e86d4b3917c6

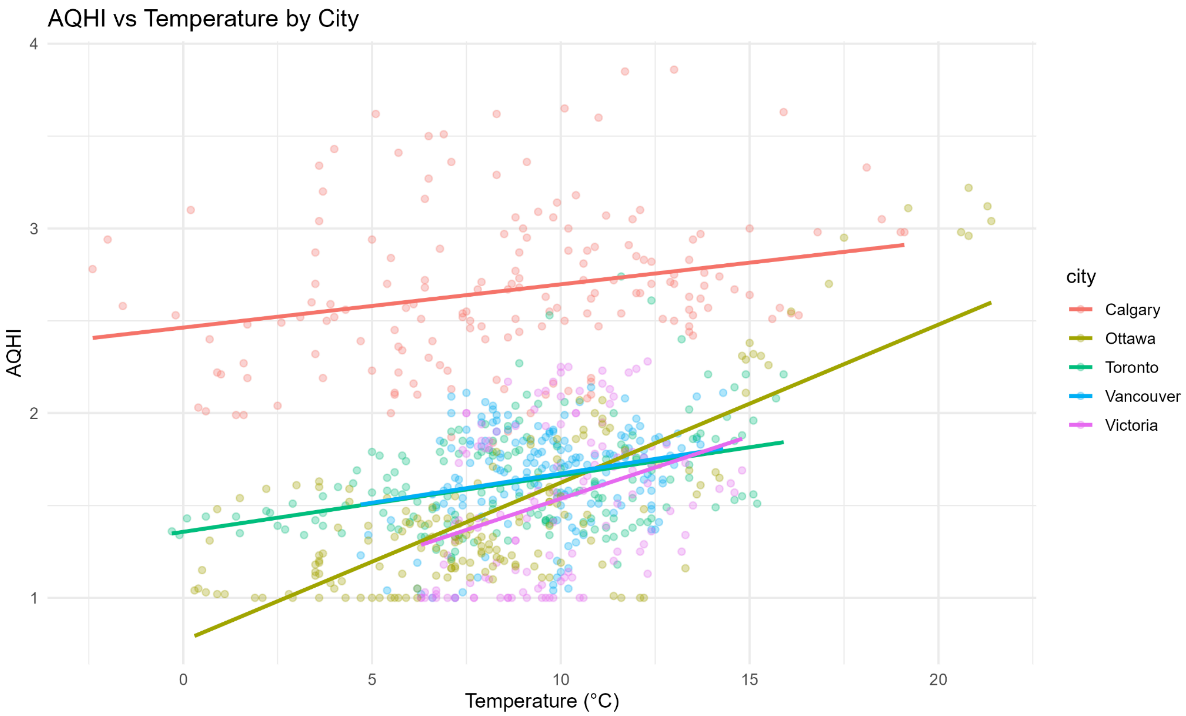
Open-Meteo. (2025). Weather Forecast API. Retrieved from https://open-meteo.com/en/docs

Appendix A - Q1 and Q1 follow up (Weather Condition)

A.1 Regression output

```
Call:
lm(formula = AQHI ~ temperature_2m + relative_humidity_2m + wind_speed_10m +
    precipitation, data = merged_inner)

Residuals:
    Min      1Q  Median      3Q     Max
-0.9347 -0.2952 -0.0184  0.2613  1.3886

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)           3.0985629  0.0862180  35.939  < 2e-16 ***
temperature_2m        0.0206044  0.0046174   4.462 9.27e-06 ***
relative_humidity_2m -0.0197690  0.0008247 -23.970  < 2e-16 ***
wind_speed_10m       -0.0032770  0.0031969  -1.025   0.3056
precipitation         0.0889695  0.0382309   2.327   0.0202 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4218 on 797 degrees of freedom
  (38 observations deleted due to missingness)
Multiple R-squared:  0.4844,    Adjusted R-squared:  0.4819
F-statistic: 187.2 on 4 and 797 DF,  p-value: < 2.2e-16
```

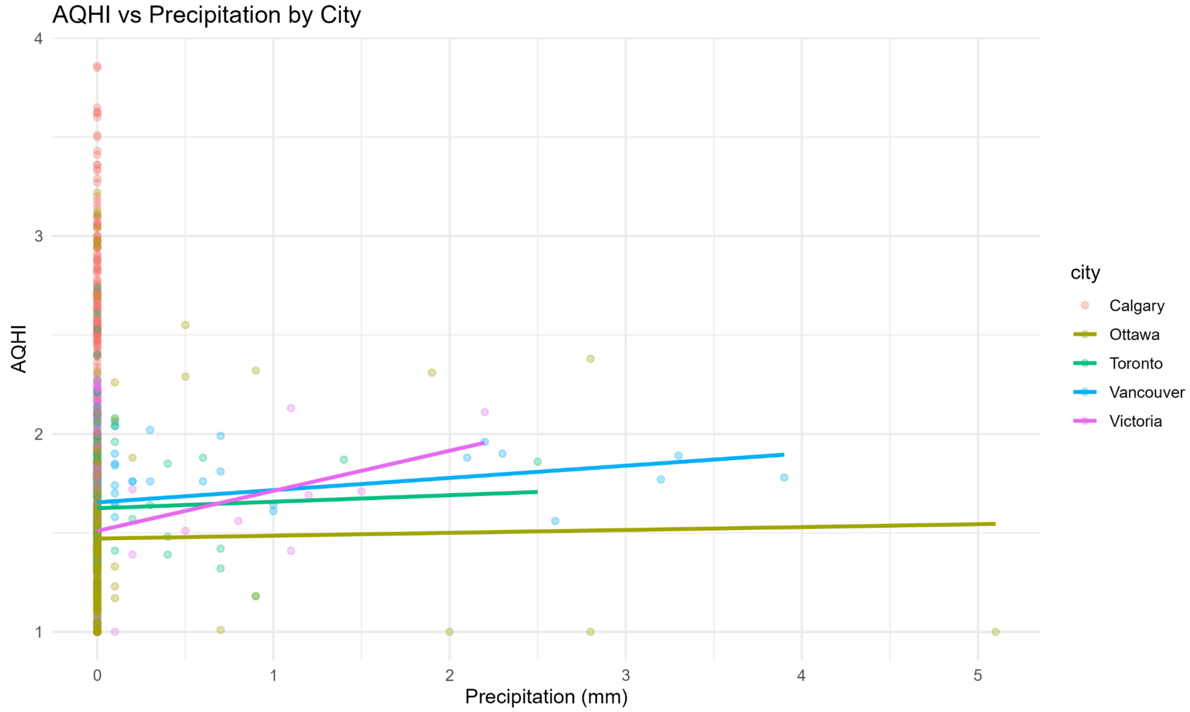
A.2 Scatterplot AQHI vs Temperature



A.3 Wind speed chart

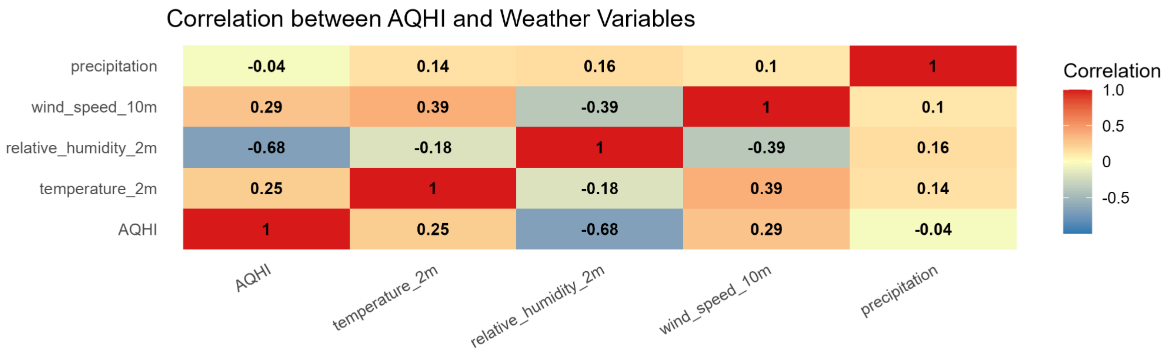
A.4 Humidity chart

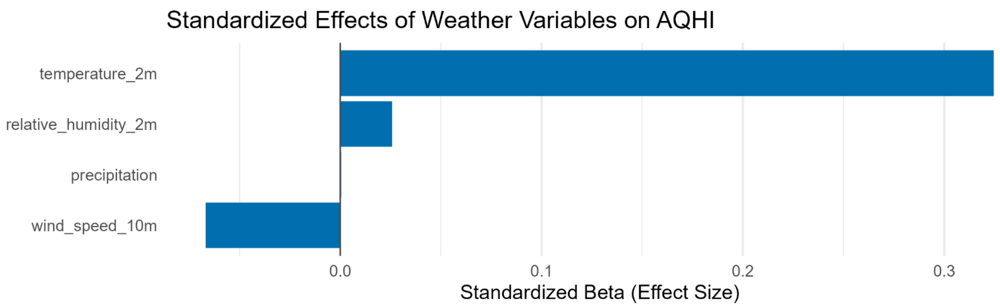
A.5 Precipitation chart



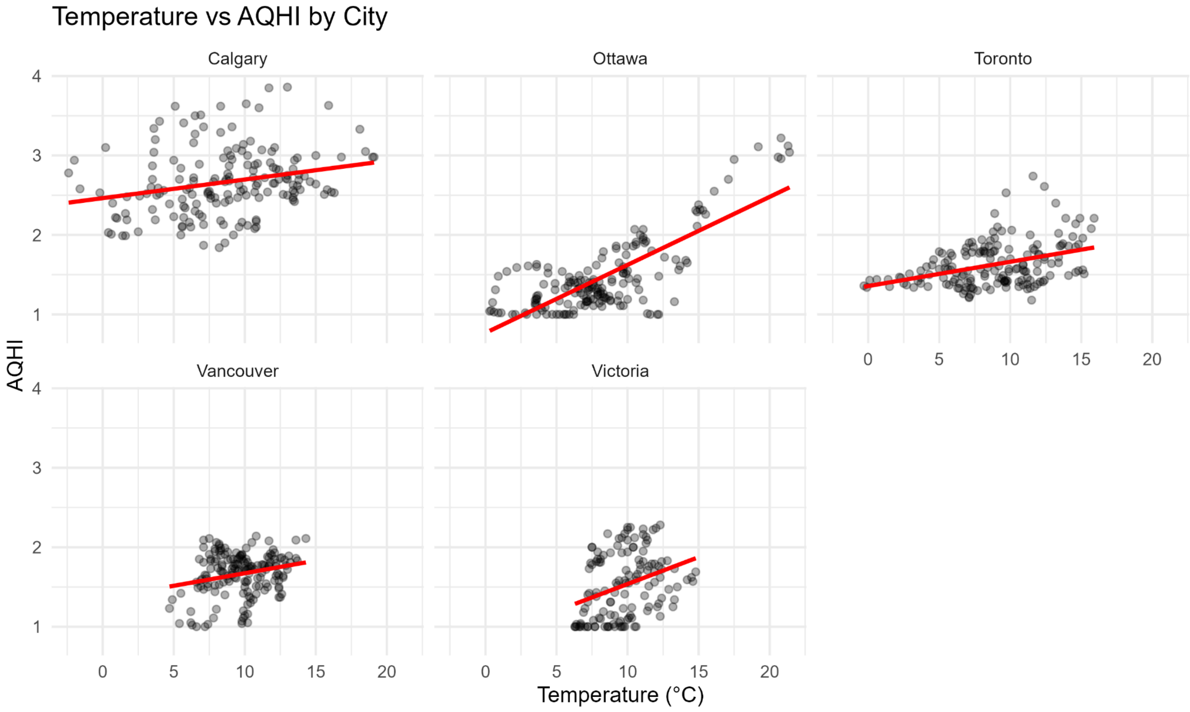
A.6 Correlation heatmap

Follow-up Question 1 – Which Weather Factor Influences AQHI Most?

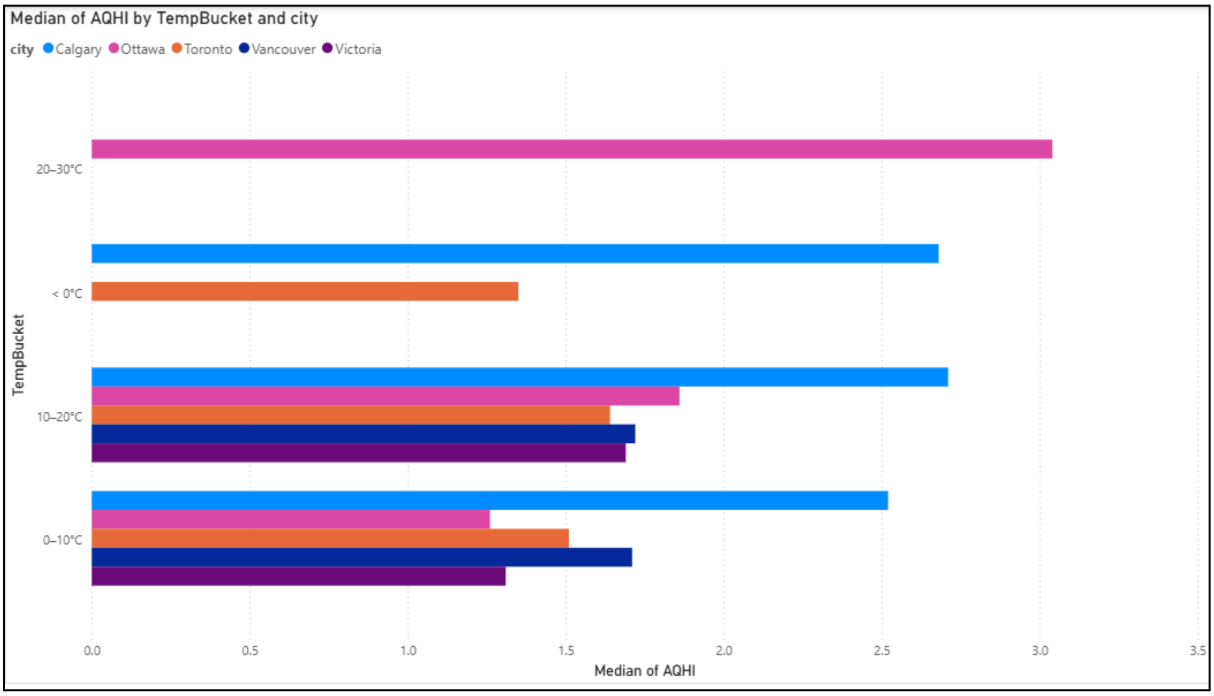




A.7 Temperature vs AQHI by City (Scatterplots with Regression Lines)



A.8 Median AQHI Across Temperature Buckets by City



Appendix B - Q2 and Q2 follow up (Location Condition)

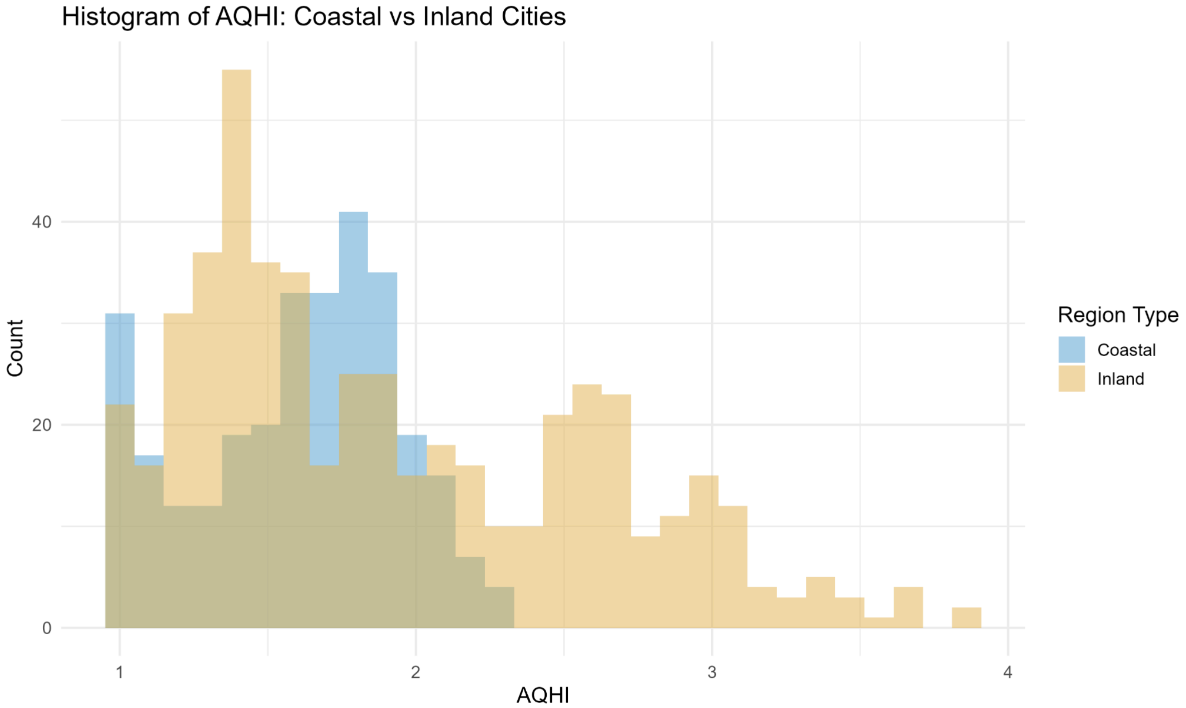
B.1 Welch's Two-Sample t-test: Coastal vs Inland Regions

```
        Welch Two Sample t-test

data:  AQHI by region_type
t = -8.9682, df = 785.98, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Coastal and group Inland is not equal to 0
95 percent confidence interval:
 -0.3883052 -0.2488438
sample estimates:
mean in group Coastal  mean in group Inland
             1.602517              1.921091

> tidy_t <- broom::tidy(t_test_result)
> print(tidy_t)
# A tibble: 1 × 10
  estimate estimate1 estimate2 statistic  p.value parameter conf.low conf.high method          alternative
     <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>    <dbl>     <dbl> <chr>           <chr>
1   -0.319      1.60      1.92     -8.97 2.17e-18      786.   -0.388    -0.249 Welch Two Samp… two.sided
> |
```

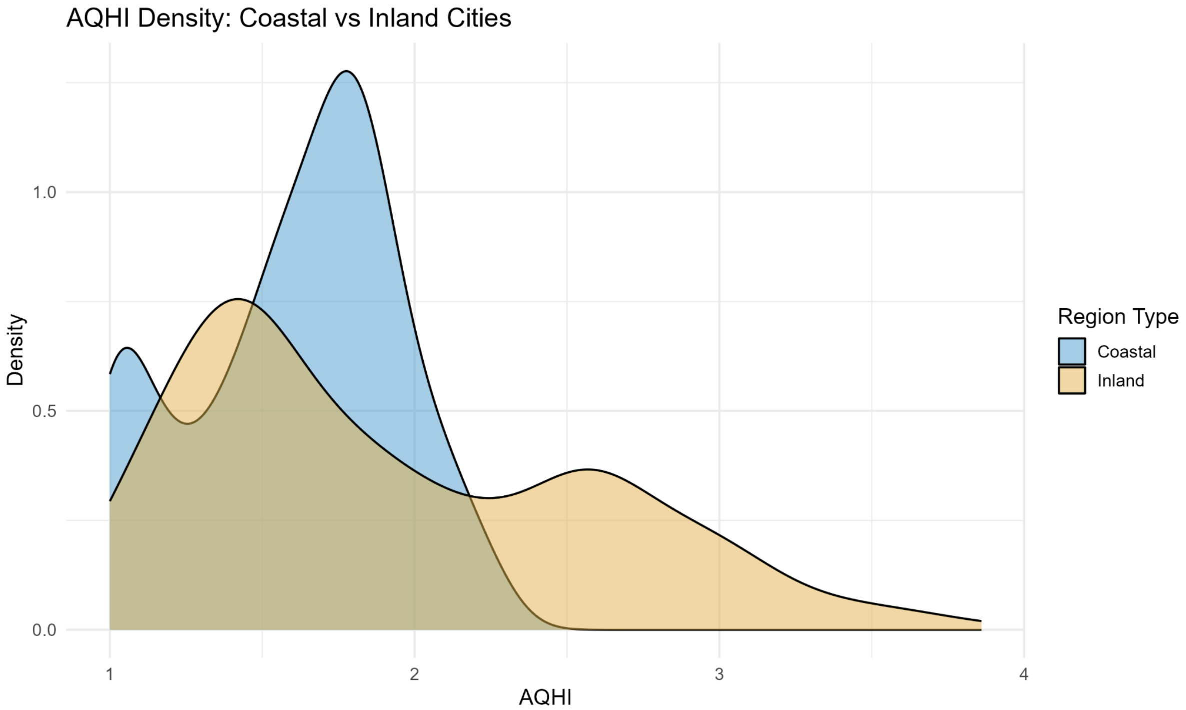
B.2 Linear Regression: Effect of Region Type (Coastal vs Inland) on AQHI

```
Call:
lm(formula = AQHI ~ region_type, data = df2)

Residuals:
    Min      1Q  Median      3Q     Max
-0.9211 -0.4811 -0.0468  0.3375  1.9389

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)        1.60252    0.03277  48.900  < 2e-16 ***
region_typeInland  0.31857    0.04134   7.706 3.84e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5657 on 800 degrees of freedom
Multiple R-squared:  0.0691,    Adjusted R-squared:  0.06794
F-statistic: 59.39 on 1 and 800 DF,  p-value: 3.837e-14

> broom::tidy(m2a)
# A tibble: 2 × 5
  term              estimate std.error statistic  p.value
  <chr>                <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)          1.60     0.0328     48.9  1.47e-242
2 region_typeInland    0.319    0.0413      7.71 3.84e- 14
> |
```

B.3 Histogram of AQHI: Coastal vs Inland Cities



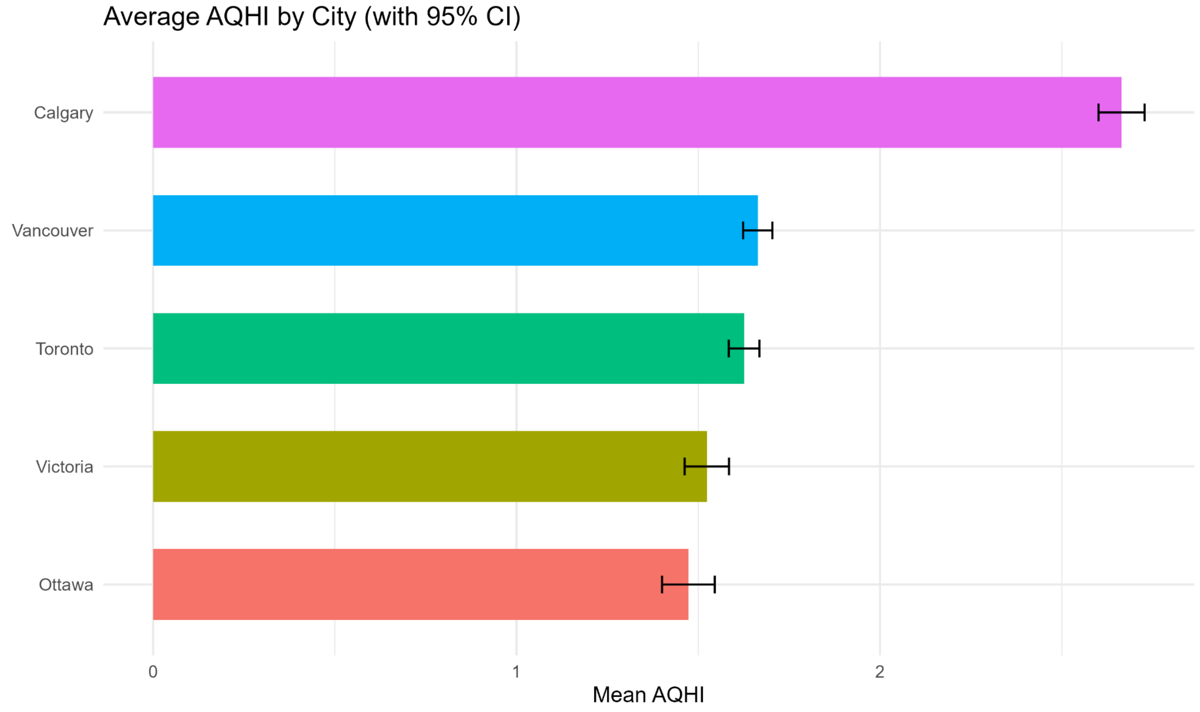
B.4 Density Plot of AQHI: Coastal vs Inland Cities



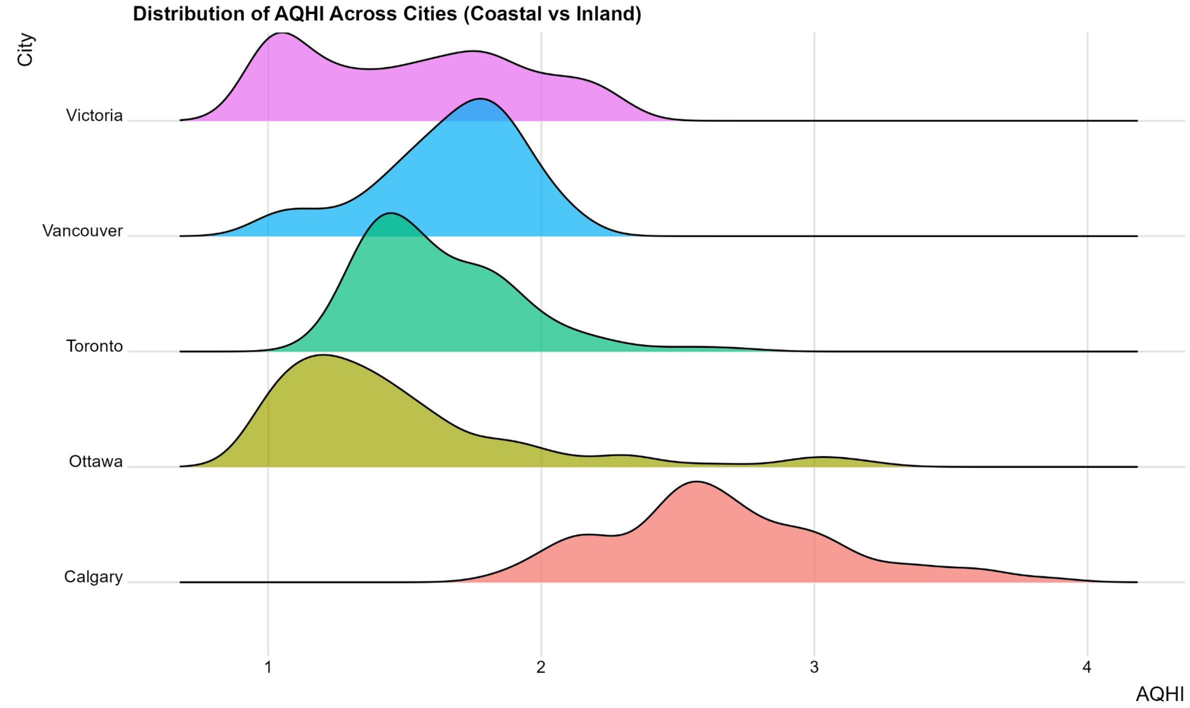
B.5 Linear Regression: AQHI Differences Across Individual Cities

```
Call:
lm(formula = AQHI ~ city, data = merged_inner)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.82440 -0.25421 -0.04402  0.20560  1.74720 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)    2.66440    0.02910   91.56   <2e-16 ***
cityOttawa    -1.19161    0.04115  -28.95   <2e-16 ***
cityToronto   -1.03833    0.04115  -25.23   <2e-16 ***
cityVancouver -1.00077    0.04115  -24.32   <2e-16 ***
cityVictoria  -1.14087    0.04406  -25.89   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3772 on 797 degrees of freedom
Multiple R-squared:  0.5877,	Adjusted R-squared:  0.5857 
F-statistic: 284.1 on 4 and 797 DF,  p-value: < 2.2e-16

> 
> broom::tidy(m_city)
# A tibble: 5 × 5
  term          estimate std.error statistic   p.value
  <chr>            <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)       2.66    0.0291      91.6 0        
2 cityOttawa       -1.19    0.0412     -29.0 1.58e-126
3 cityToronto      -1.04    0.0412     -25.2 1.06e-103
4 cityVancouver    -1.00    0.0412     -24.3 3.80e- 98
5 cityVictoria     -1.14    0.0441     -25.9 9.32e-108
> 
```

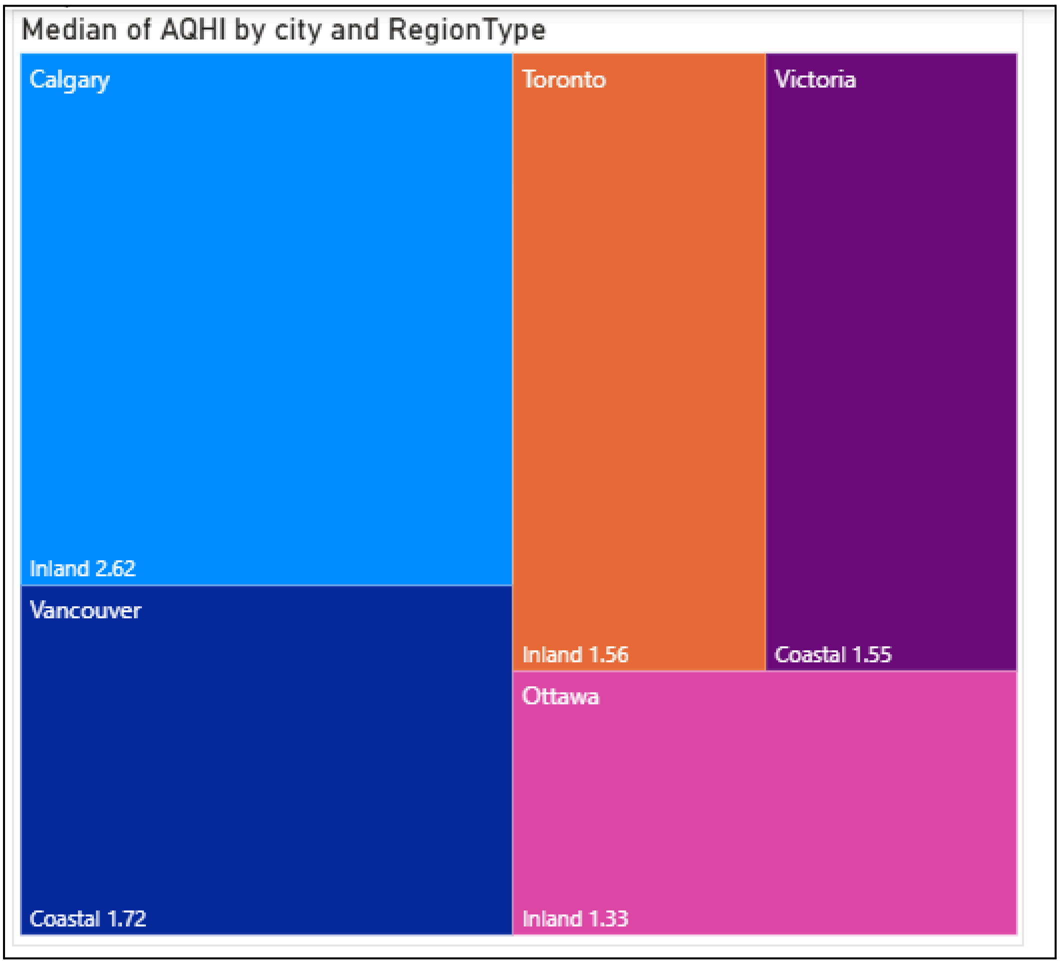
B.6 Average AQHI by City with 95% Confidence Intervals



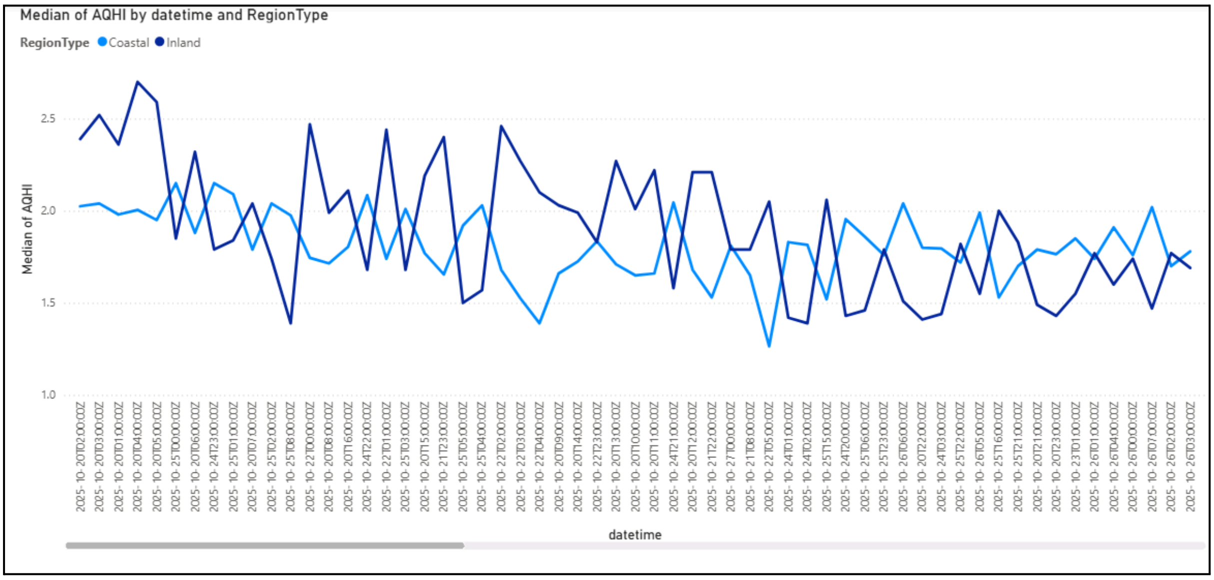
B.7 Density Distribution of AQHI Across Cities (Coastal vs Inland Grouping)



B.8 Treemap of Median AQHI by City and Region Type



B.9 Time-Series Plot of Median AQHI by Region Type (Coastal vs Inland)



Appendix C - Q3 and Q3 follow up (Time Condition)

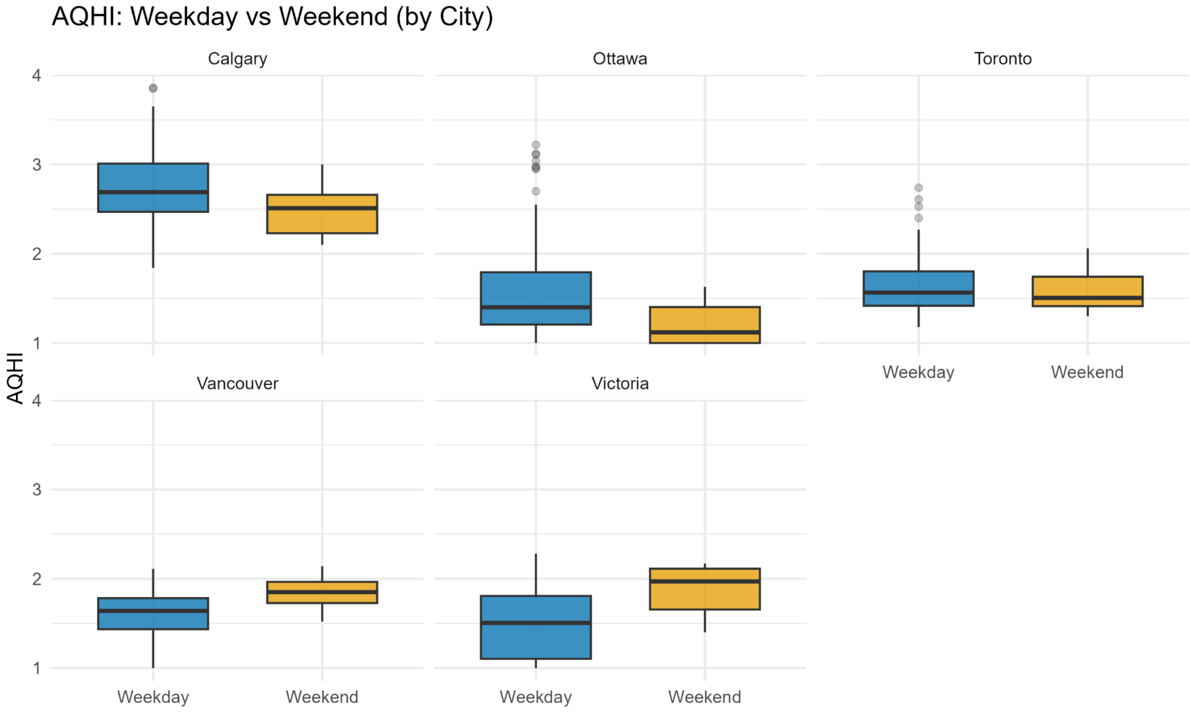
C.1 T-Test Results: Weekday vs Weekend (Overall and by City)

```
> tt_overall <- t.test(AQHI ~ day_type, data = df3)
> print(tt_overall %>% tidy())
# A tibble: 1 × 10
  estimate estimate1 estimate2 statistic p.value parameter conf.low conf.high method                  alternative
     <dbl>     <dbl>     <dbl>     <dbl>   <dbl>     <dbl>    <dbl>     <dbl> <chr>                   <chr>
1   0.0247      1.81      1.78     0.571   0.569      422.  -0.0604     0.110 Welch Two Sample t-test two.sided
>
>
> tt_city <- df3 %>%
+   group_by(city) %>%
+   do(tidy(t.test(AQHI ~ day_type, data = .))) %>%
+   ungroup()
>
> print(tt_city)
# A tibble: 5 × 11
  city      estimate estimate1 estimate2 statistic  p.value parameter conf.low conf.high method                  alternative
  <chr>        <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>    <dbl>     <dbl> <chr>                   <chr>
1 Calgary     0.244       2.73      2.49      4.41 2.02e- 5     146.   0.135       0.354 Welch Two Sample t-test two.sided
2 Ottawa      0.366       1.58      1.21      6.48 1.02e- 9     165.   0.254       0.477 Welch Two Sample t-test two.sided
3 Toronto     0.0753      1.65      1.57      1.87 6.40e- 2     127.  -0.00446     0.155 Welch Two Sample t-test two.sided
4 Vancouver  -0.252       1.59      1.84     -7.48 7.98e-12     138.  -0.319      -0.186 Welch Two Sample t-test two.sided
5 Victoria   -0.373       1.49      1.86     -3.94 1.28e- 3      15.1 -0.575      -0.172 Welch Two Sample t-test two.sided
>
```

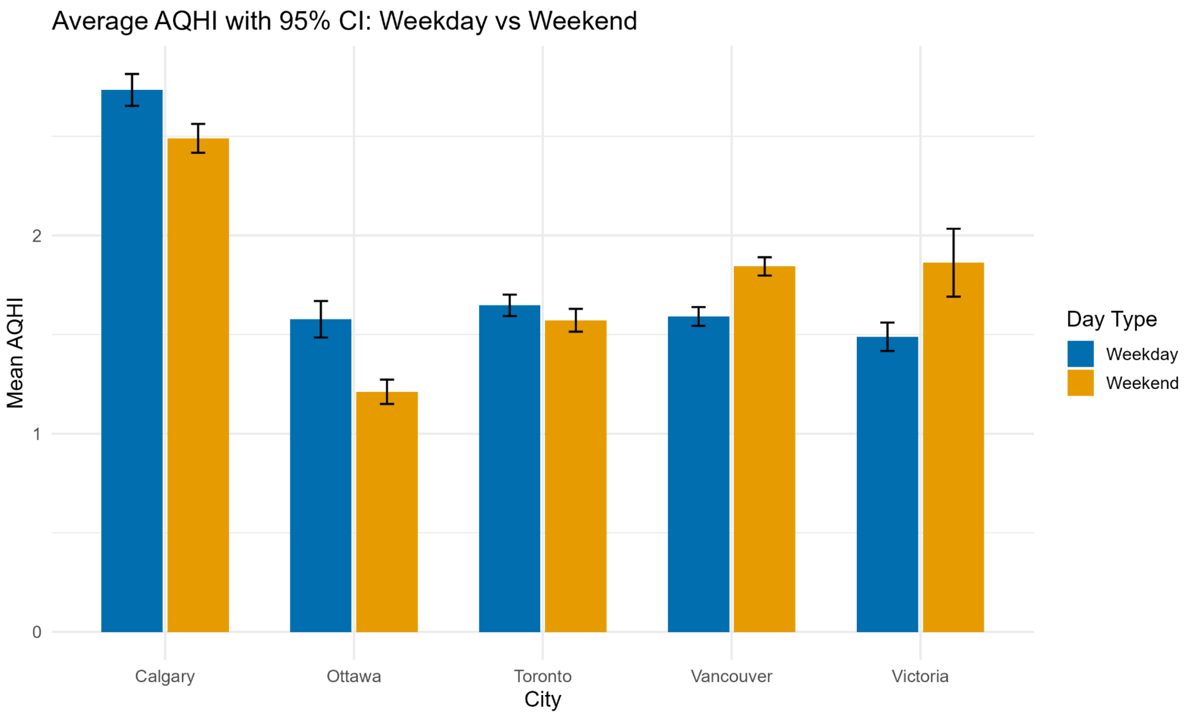
C.2 Regression Output: AQHI Predicted by Day Type

```
Call:
lm(formula = AQHI ~ day_type, data = df3)

Residuals:
    Min      1Q  Median      3Q     Max 
-0.8090 -0.4190 -0.0990  0.2985  2.0510 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)      1.80900    0.02397  75.459   <2e-16 ***
day_typeWeekend -0.02468    0.04753  -0.519    0.604    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.5862 on 800 degrees of freedom
Multiple R-squared:  0.0003369,	Adjusted R-squared:  -0.0009126 
F-statistic: 0.2696 on 1 and 800 DF,  p-value: 0.6037


>
> tidy_m3_overall <- broom::tidy(m3_overall)
> print(tidy_m3_overall)
# A tibble: 2 × 5
  term            estimate std.error statistic p.value
  <chr>              <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)       1.81      0.0240     75.5    0    
2 day_typeWeekend  -0.0247    0.0475     -0.519  0.604
```

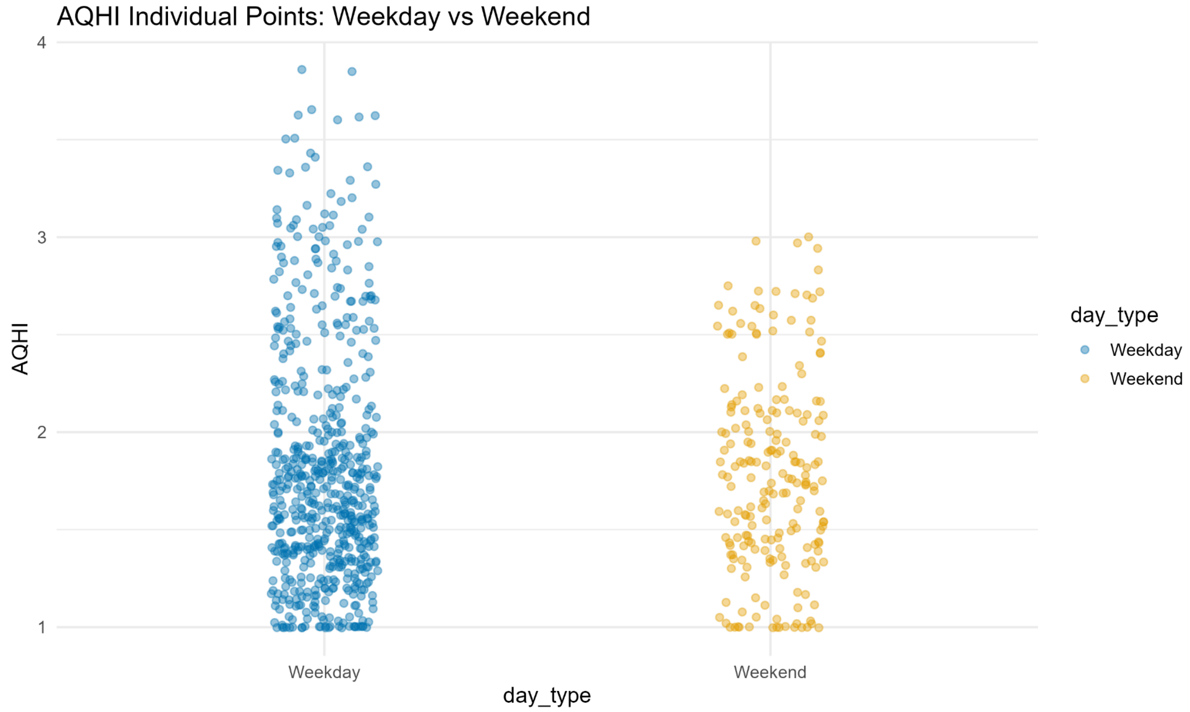
C.3 Boxplots of AQHI: Weekday vs Weekend by City



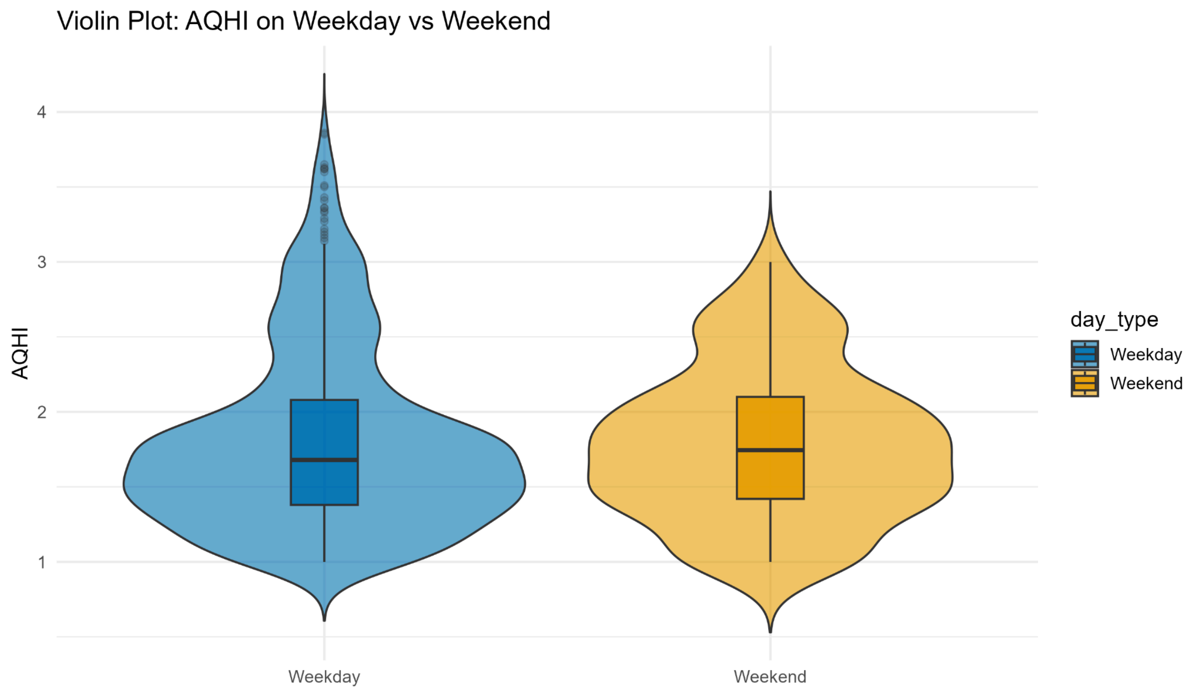
C.4 Mean AQHI with 95% CI: Weekday vs Weekend



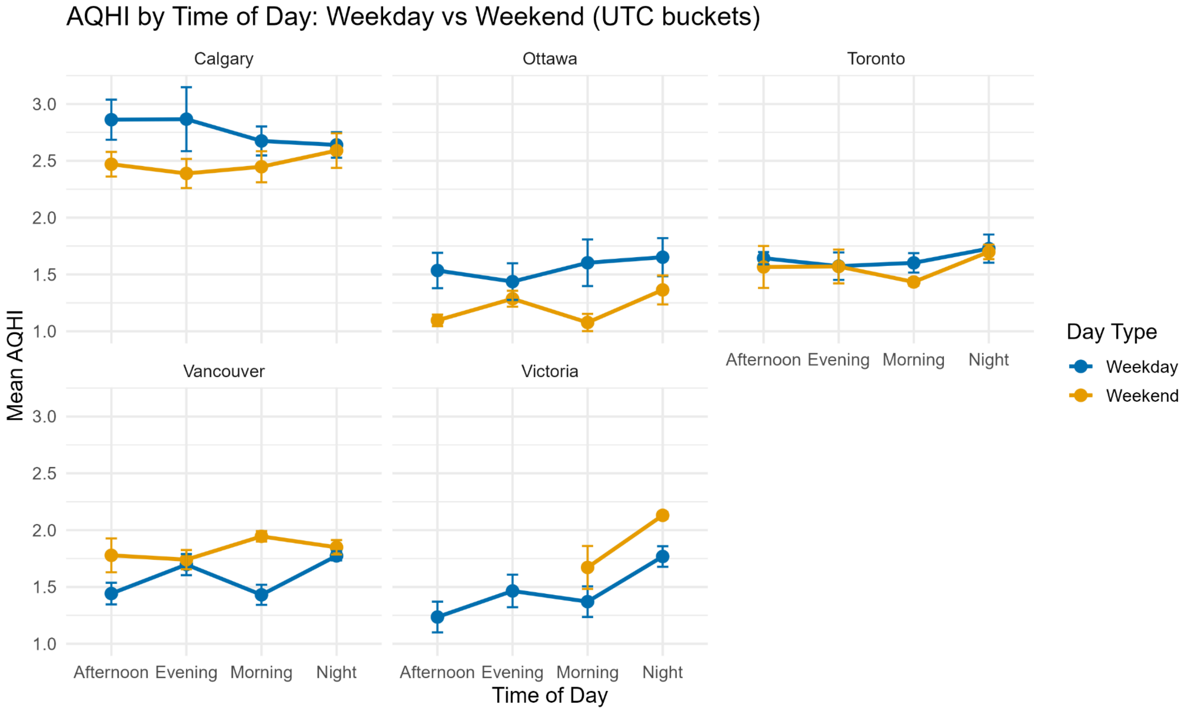
C.5 Scatter Plot of Individual AQHI Points (Weekday vs Weekend)



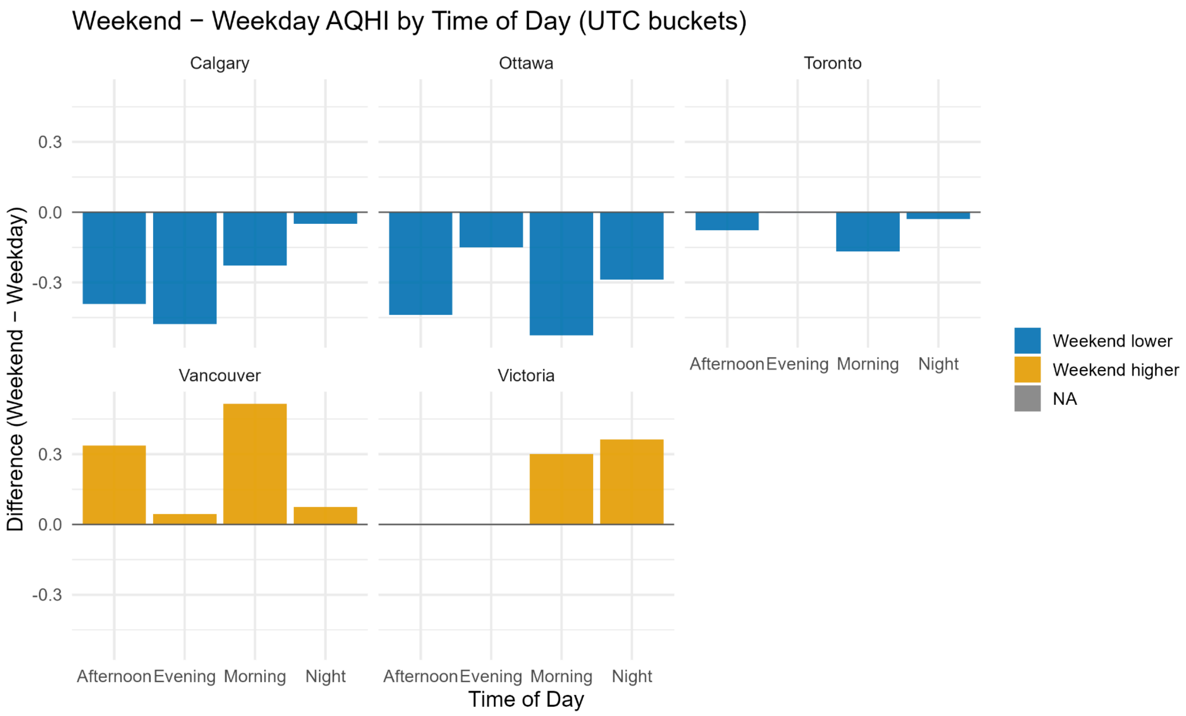
C.6 Violin Plot of AQHI: Weekday vs Weekend



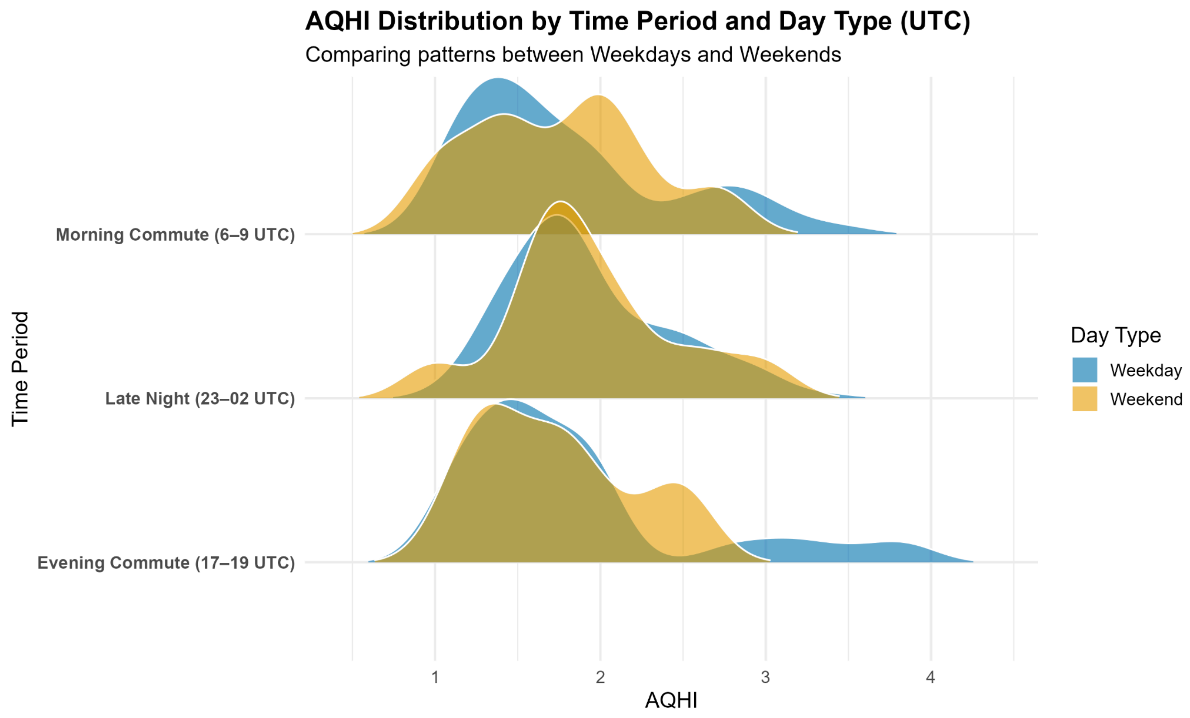
C.7 Line Chart: AQHI by Time of Day (Weekday vs Weekend, by City)



C.8 Difference Plot (Weekend – Weekday) by Time of Day and City



C.9 Ridgeline Density Plot: AQHI by Time Period and Day Type



C.10 Circular Chart: Mean AQHI by Time of Day